COMPUTER VISION WITH DEEP LEARNING · FINAL PROJECT

TriFrame CV

Can pictures of stock charts predict risk?

500 US stocks · 236,520 chart sets · 7 pretrained vision networks · 2001–2022



Krishna Harsha Tangirala · github.com/harshatangirala/triframe-cv · October 2026

THE ANSWER IN ONE SLIDE

Risk is predictable. Direction is not.

20-DAY VOLATILITY

0.503

R^2 on the 2020–2022 test years

Explains half the variation in future volatility, ahead of classical models on squared error.

10% DRAWDOWN

0.648

ROC-AUC on the test years

Ranks a stock that later falls 10% above one that does not 65% of the time.

5-DAY DIRECTION

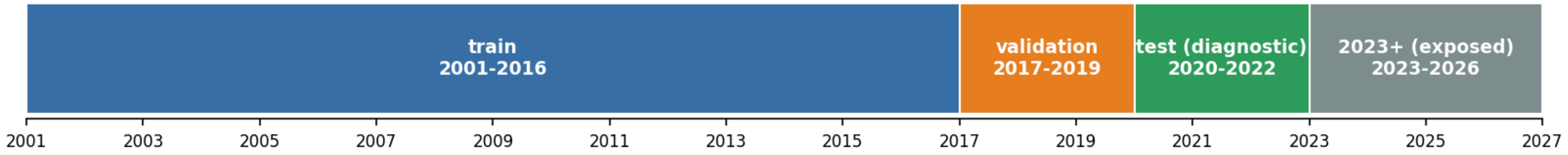
0.510

ROC-AUC on the test years

A coin flip. The decision layer therefore defaults to NO TRADE.

Chart images add no significant gain in overall R^2 on the test years: +0.003 (95% CI −0.018 to +0.026). A later test finds a gain within months: +0.020 (95% CI 0.012 to 0.027).

236,520 chart sets from 500 US stocks



Period	Years	Chart sets	Stocks	10% drawdowns
Train	2001–2016	165,421	489	22.7%
Validation	2017–2019	35,145	492	17.8%
Test	2020–2022	35,954	500	30.2%
2023 onward	exposed	–	–	unused

979
US stocks with daily and hourly OHLCV bars

500
sampled at random; they contain the 60- and 120-stock runs

1,913
rows purged where a 20-day label crossed a split

Three text-free charts per stock-day



Renderer

OHLC bars, 3 px each, 128 px high: a 100 px price panel over a 26 px volume panel. No text, no gridlines.

Network input

Pixels copied to 3 channels, resized to 224 × 224 with antialiasing and normalised with each network's ImageNet statistics.

The key consequence

Log prices are stretched to fill each panel, so a picture shows the shape of moves, never their size.

Three targets, 33 numbers, one window

V **20-day volatility**

Log realised volatility of daily returns over the next 20 sessions

`regression · $R^2$, RMSE`

D10 **10% drawdown**

Price trades 10% or more below the next day's open within 20 sessions

`yes/no · base rate 22.7 / 17.8 / 30.2%`

05 **Beats the median stock**

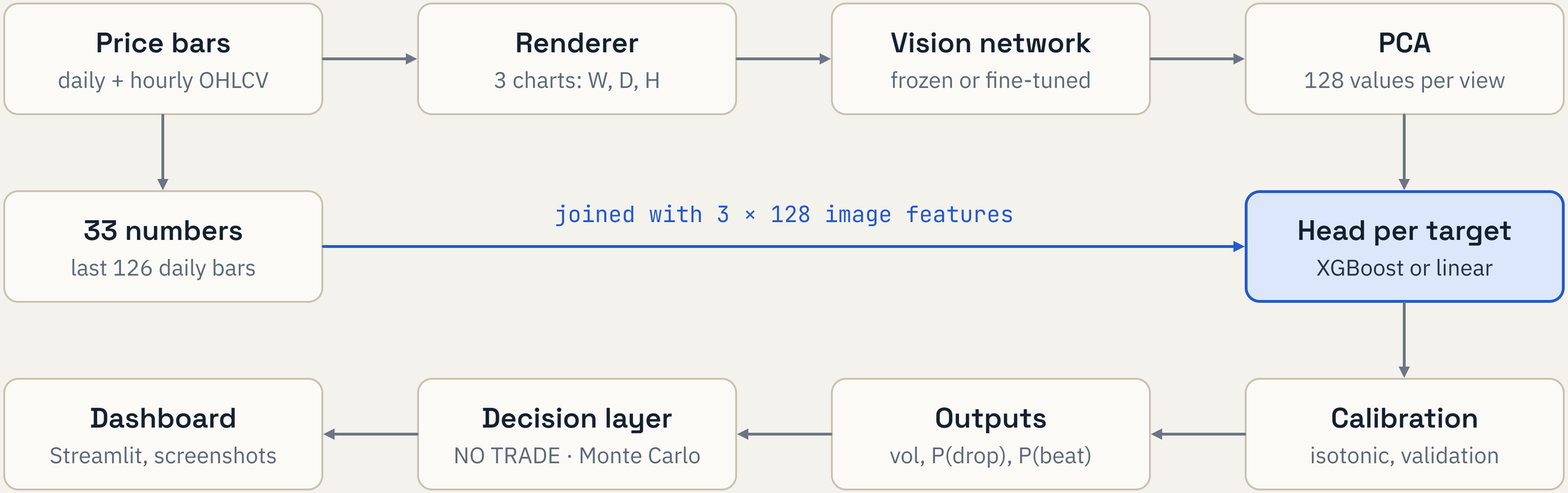
5-day return above the same-date median of the universe

`yes/no · base rate 50%`

Group	The 33 numeric features
Returns	1-, 5- and 21-day returns; overnight vs intraday split; gap
Volume	abnormal volume; 5-day volume surge
Mean reversion	RSI (2, 3, 14); 20-day z-score; streaks; variance ratio; AR(1)
Volatility	close-to-close, Parkinson and Garman-Klass at 1, 5 and 22 days; EWMA
Amplitude	ATR %, log range, largest drawdown in the window

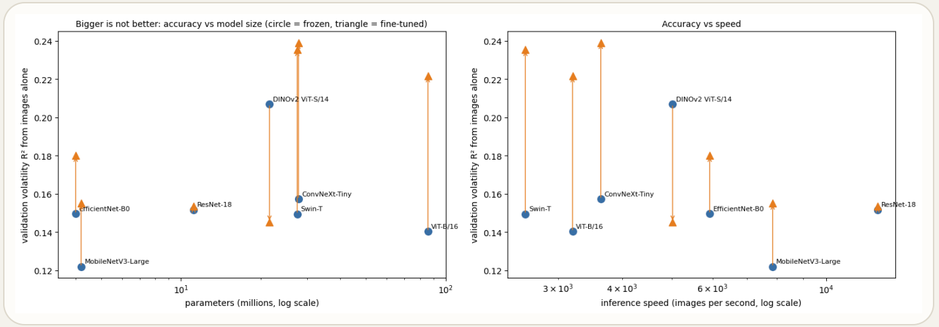
All 33 come from the last 126 daily bars, the same window a daily screenshot shows. Gaps are filled with training medians; scaling is fitted on training years only.

Images and numbers meet in one head



Seven networks, three random controls

Network	Type	Pretraining data	Params
ViT-B/16	Transformer	ImageNet-21k → 1k	85.8 M
ConvNeXt-Tiny	CNN	ImageNet-12k → 1k	27.8 M
Swin-T	Transformer	ImageNet-1k	27.5 M
DINOv2 ViT-S/14	Transformer	LVD-142M, self-sup.	21.6 M
ResNet-18	CNN	ImageNet-1k	11.2 M
MobileNetV3-L	CNN	ImageNet-1k	4.2 M
EfficientNet-B0	CNN	ImageNet-1k	4.0 M
ResNet-18 × 3	Control	none (random)	11.2 M



Bigger is not better: R^2 from images alone against size and speed.

DINOv2 (21.6 M) reads the most from pictures alone: R^2 0.207. Random-weight ResNets: about −0.02.

Heads: ridge, logistic and XGBoost (depth 3–7, ≤ 2,000 trees). 1,881 configurations logged.

Six safeguards against fooling ourselves

Time-ordered splits

Train 2001–16, validate 2017–19, test 2020–22. 1,913 rows whose 20-day label crossed a boundary were purged.

Train-only fitting

Gap filling, scaling and PCA learn from training rows only, so nothing from later years leaks backwards.

Honest uncertainty

Overlapping 20-day labels are not independent, so every CI resamples whole calendar months (1,000 draws).

Pre-registered rule

An image model wins only if its gain over numbers has a CI above zero and holds in every validation year.

Negative controls

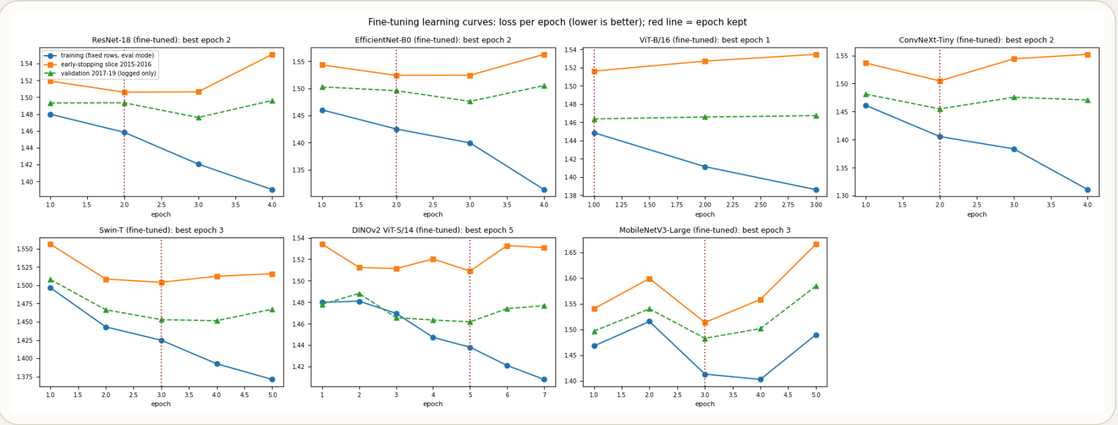
Shuffled labels give R^2 −0.123; random-weight networks about −0.02. Nothing is found where nothing exists.

Provenance

Every metric carries code and config hashes; the deployed bundle matches the notebook to within 4×10^{-5}.

End-to-end fine-tuning, validation untouched

Setting	Value
Loss	joint MSE + BCE, 3 targets
Optimiser	AdamW, weight decay 0.05
Learning rate	3e-5 backbone · 1e-3 head
Schedule	5% warm-up, cosine decay
Batch · precision	64 · bfloat16 · A100
Gradient clipping	norm 1.0
Data per epoch	fresh 60,000 of 141,525 sets
Early stopping	2015–16 slice, patience 2
Max epochs	8
Augmentation	none



Loss per epoch; the red line marks the epoch kept.

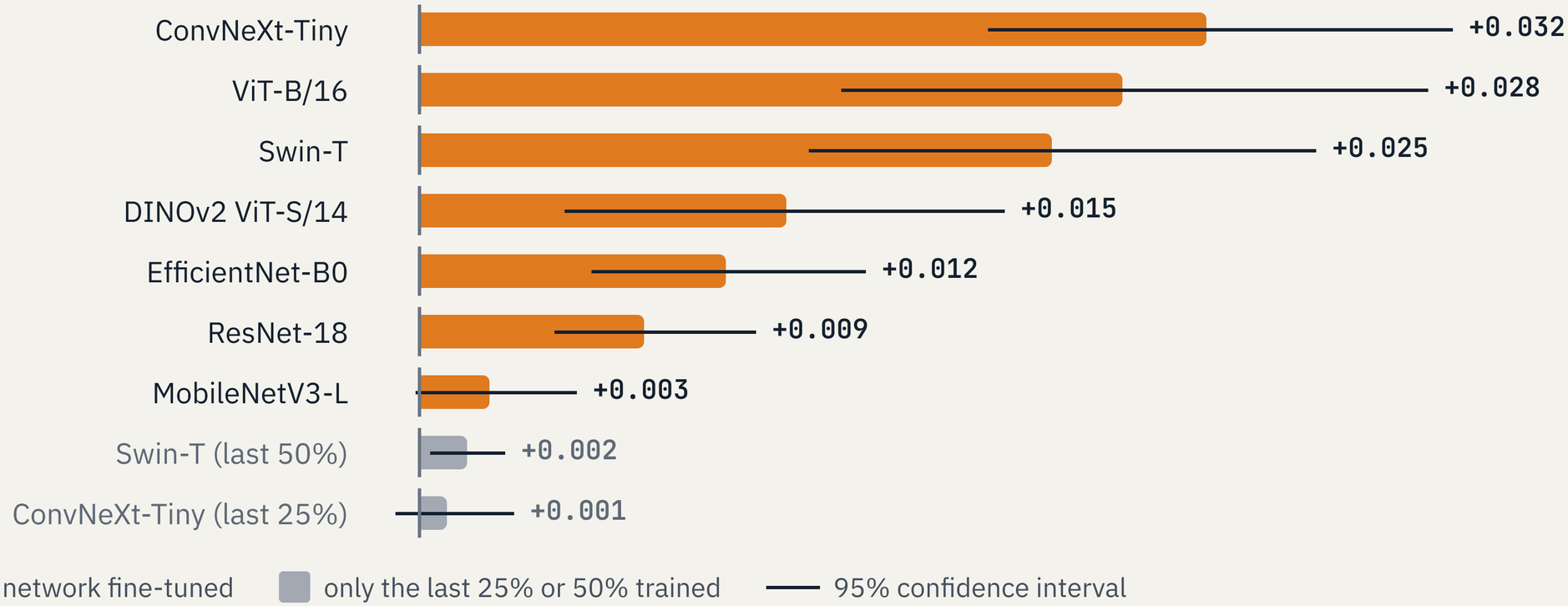
15
fine-tuning runs

158
GPU-minutes

0
validation rows used

Fine-tuning works only on the whole network

Validation R^2 gain from fine-tuning (numbers + images) with 95% month-block CIs. 9 of 9 runs improved, 7 significantly.



Convex heads: a certified global minimum

CERTIFIED GLOBAL MINIMUM

Ridge head (volatility)

Gradient at the solution 6.1×10^{-16}; 5 random starts reach it.

CERTIFIED GLOBAL MINIMUM

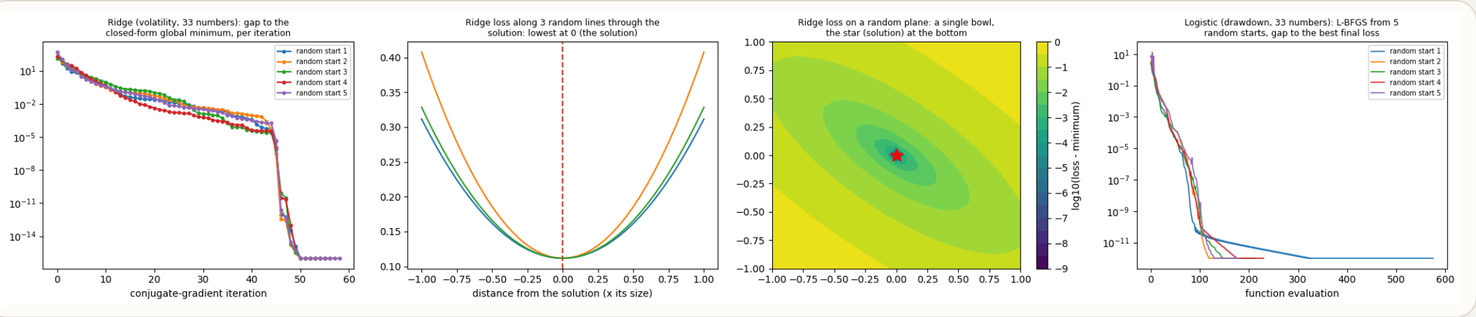
Logistic head (drawdown)

5 random starts agree in loss to 2.5×10^{-14}.

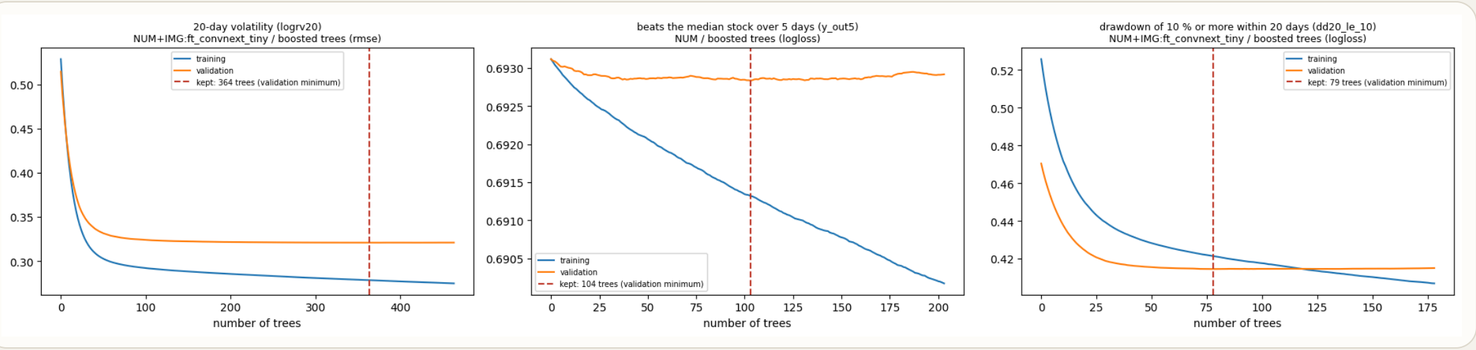
VALIDATION MINIMUM KEPT

Boosted trees

Greedy, not convex: kept at 364 (V), 79 (D10) and 104 (O5) trees.

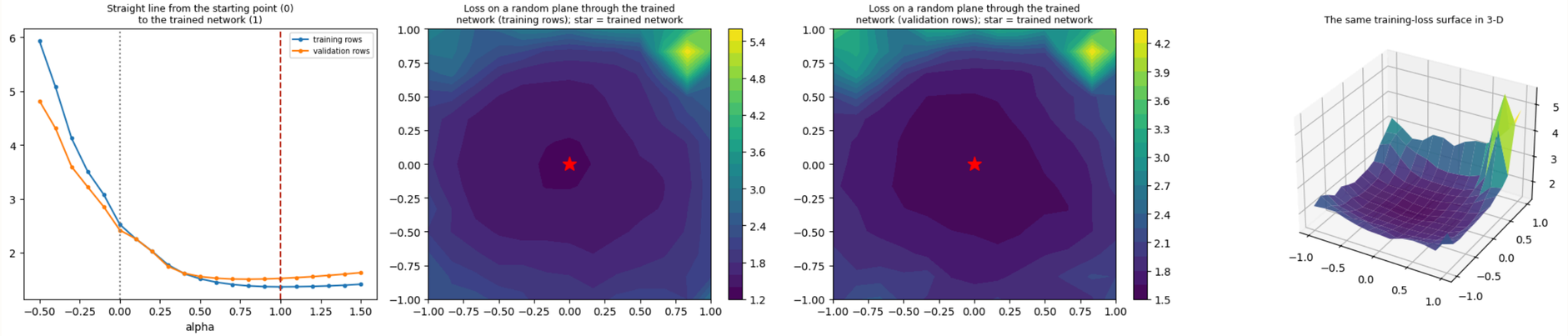


Five random starts, one minimum; the ridge loss is a single bowl.



Boosting curves: training keeps falling; the dashed line is the kept validation minimum.

The network sits at the bottom of its valley



1.461 → 1.311
training loss over 4 epochs

Epoch 2 of 4
lowest held-out loss; that checkpoint is kept

α = 1.0
lowest point on the start-to-trained path

0.007
seed-to-seed SD of validation R^2

No method can certify a global minimum for a non-convex network. The target is the lowest held-out loss, not the lowest training loss, which would only memorise the training years.

Scores on the 2020–2022 test years

Target	Metric	Validation	Test (95% CI)	Numbers only, test
Volatility	R^2	0.592	0.503 (0.25–0.65)	0.500
Volatility	RMSE, log vol	0.321	0.381	0.382
Volatility	Spearman ρ	0.758	0.742	0.726
Drawdown	ROC-AUC	0.740	0.648 (0.59–0.69)	0.653
Drawdown	Prec. / recall / F1	0.36 / 0.55 / 0.43	0.39 / 0.65 / 0.49	0.37 / 0.74 / 0.49
Direction	ROC-AUC	0.516	0.510 (0.500–0.521)	0.510

THRESHOLDS

Fixed on validation: 0.223 for drawdown, 0.439 for direction.

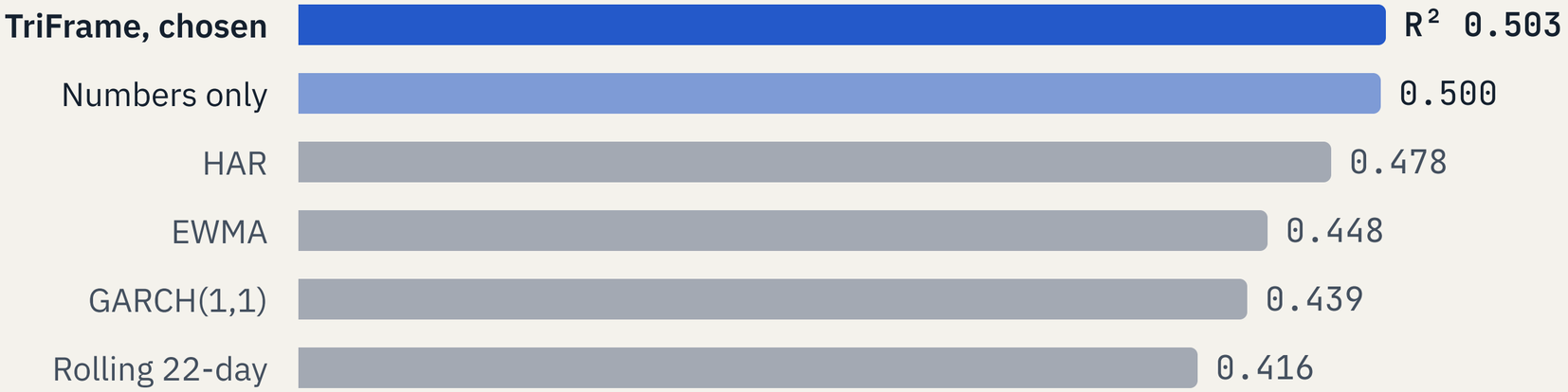
DRAWDOWN, TEST

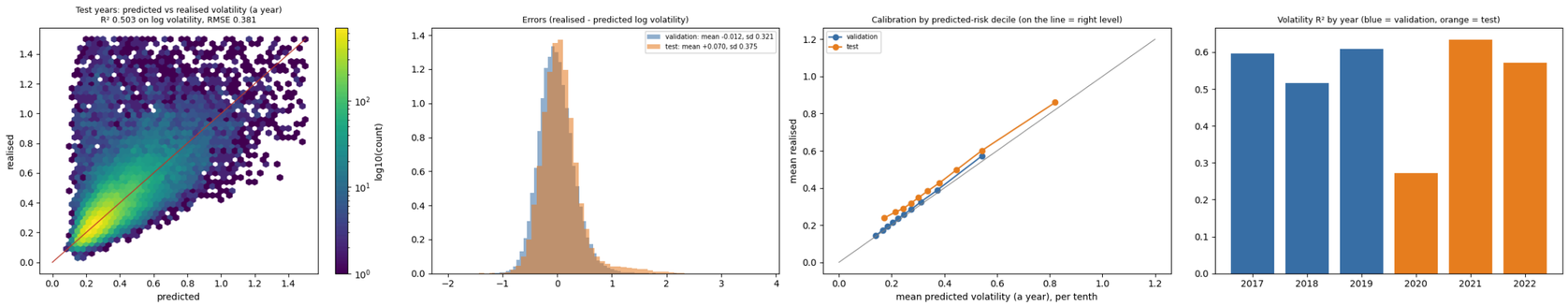
Balanced accuracy 0.607; Brier skill 0.046 over the base rate.

STATUS OF THE TEST YEARS

Read by earlier runs, so a historical check, not a fresh test.

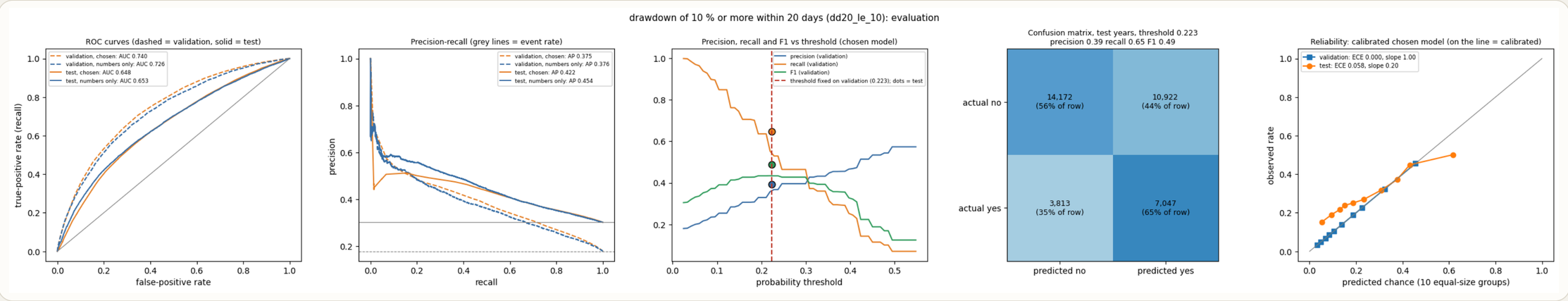
Ahead of classical models on squared error





Test R². The gain over GARCH is significant; the gain over HAR is not; under QLIKE the simple models win. Below: predicted vs realised, errors, calibration and R² by year.

Drawdown risk ranked well above chance



ROC and precision-recall curves, threshold curves, test confusion matrix and reliability (dashed = validation, solid = test).

0.648
ROC-AUC, 95% CI 0.59 to 0.69

0.39
precision against a 0.30 base rate: 1.3× lift

0.65
recall at the validation threshold of 0.223

0.058
calibration error: base rate rose from 17.8% to 30.2%

The risk flag separates risky stocks

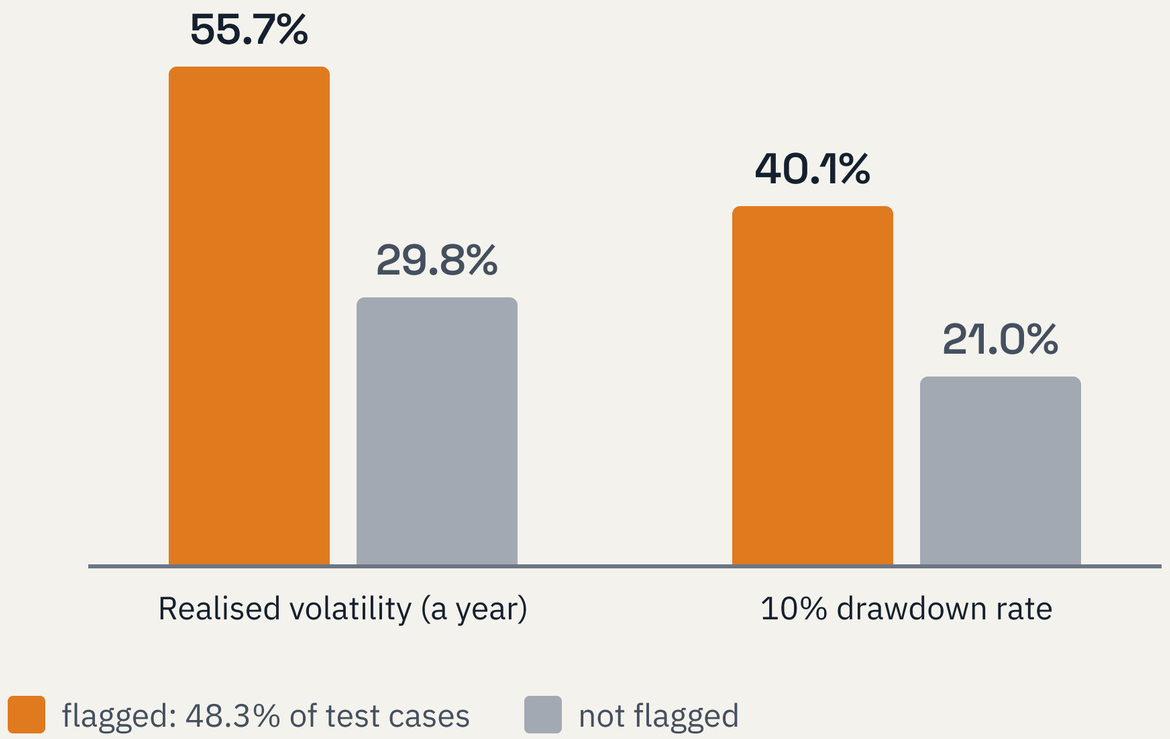
RULE FIXED ON VALIDATION

Flag a stock as high risk when

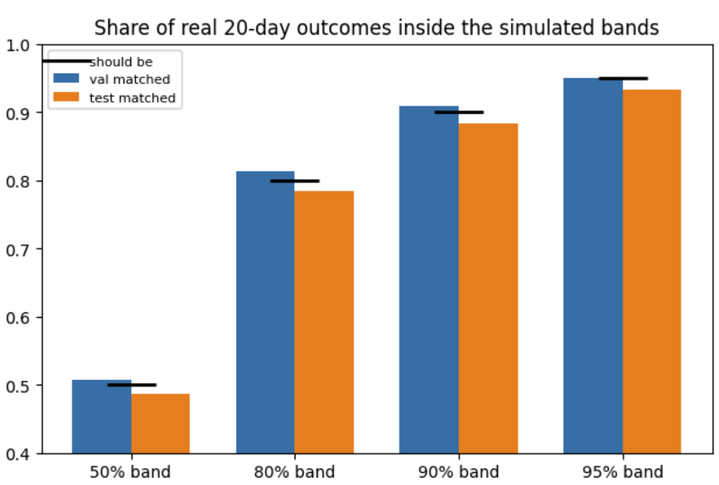
- predicted volatility ≥ 33.6% a year
- or P(10% drawdown) ≥ 30.3%

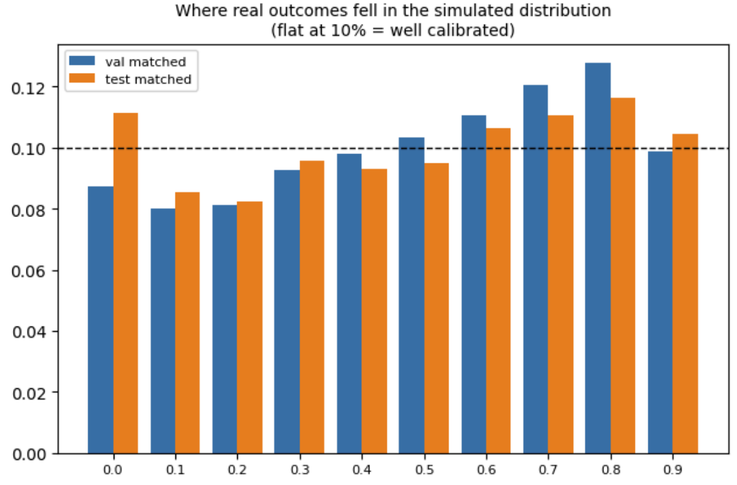
Both cut-offs are the top fifth of validation readings.

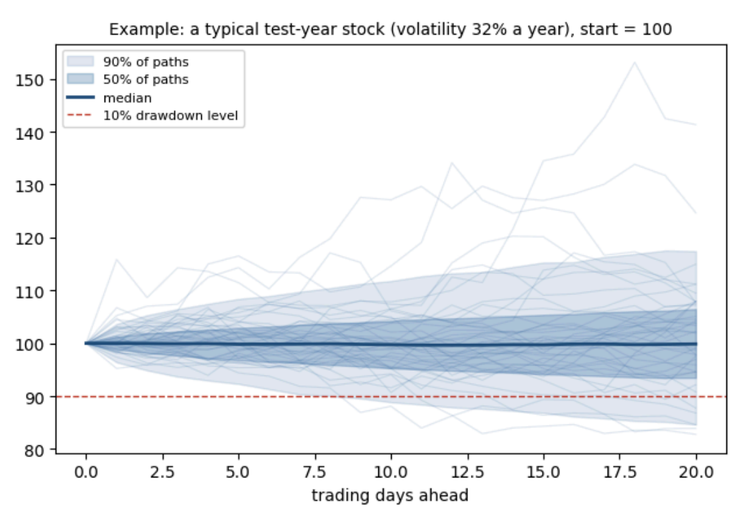
A direction call appears only if direction passes both validation and test checks. It did not, so every decision is **NO TRADE**.



Simulated price paths, calibrated on validation







Band coverage against the target, where real outcomes fell in the simulated distribution, and an example fan of 20-day paths.

88%
of test outcomes inside the 90% band (validation 91%)

49%
inside the 50% band (validation 51%)

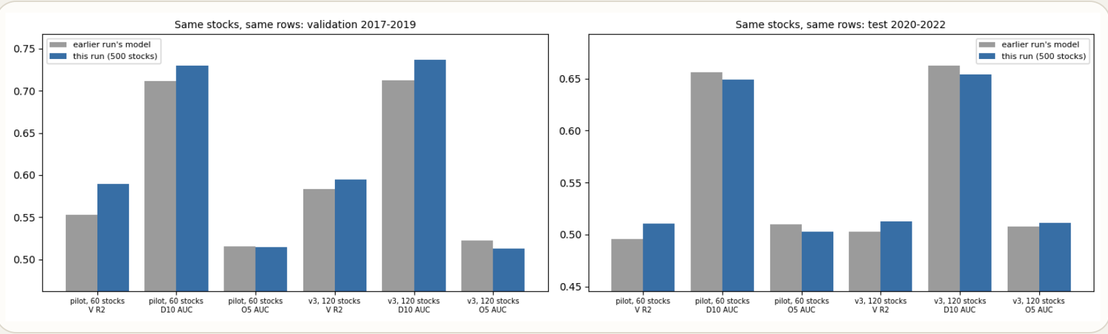
5 df
Student-t daily shocks: fat tails chosen on validation

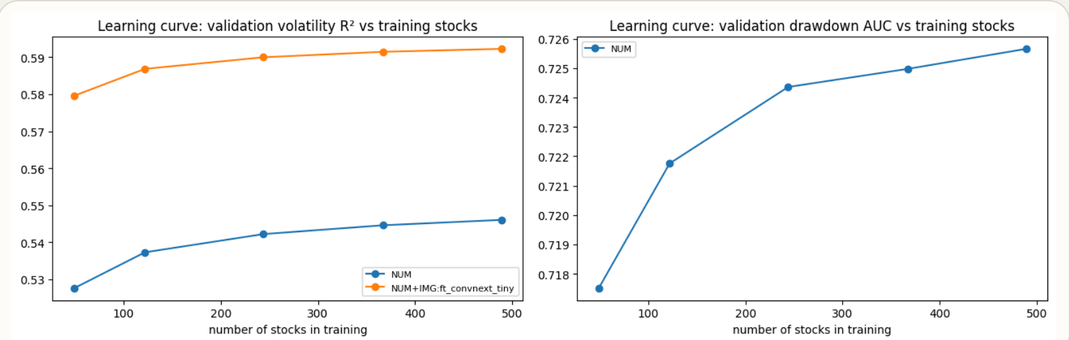
0 drift
no direction edge, so no assumed trend

More stocks helped, with diminishing returns

Earlier runs against this run, on the earlier runs' own stocks and exactly the same test rows:

Run	Metric	Earlier	500-stock	Change
Pilot (60)	Volatility R²	0.496	0.510	+0.014
v3 (120)	Volatility R²	0.503	0.513	+0.010
Pilot (60)	Drawdown AUC	0.656	0.649	−0.007
v3 (120)	Drawdown AUC	0.663	0.654	−0.009





Validation score against the number of training stocks.

Numbers-only volatility R² rose from 0.528 with 49 stocks to 0.546 with 489, and flattens after about 250. More of the same data will not move R² much further.

Images help across stocks, not in overall R²

Gain from adding the three chart views (95% CIs):

Volatility ΔR², validation: +0.046 (+0.035 to +0.059)

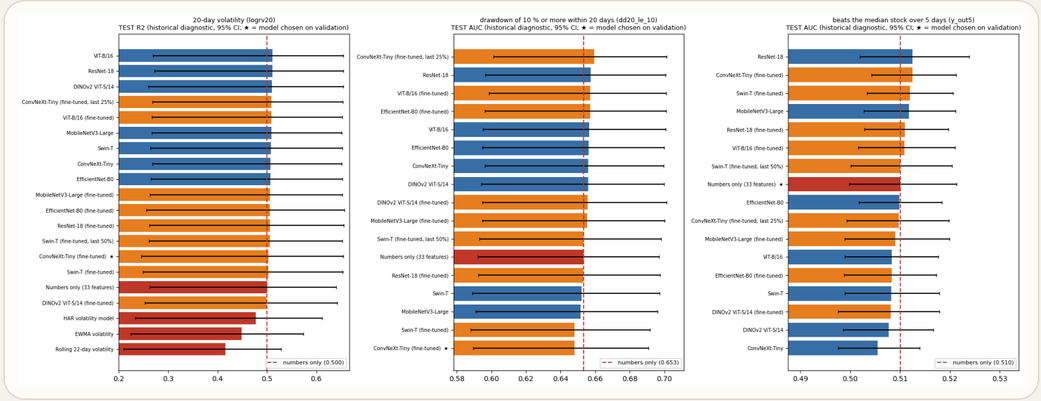
Volatility ΔR², test: +0.003 (−0.018 to +0.026)

Within-month R², test: +0.020 (+0.012 to +0.027), chosen after seeing the data

Direction ΔAUC, test: −0.002 (−0.009 to +0.005)



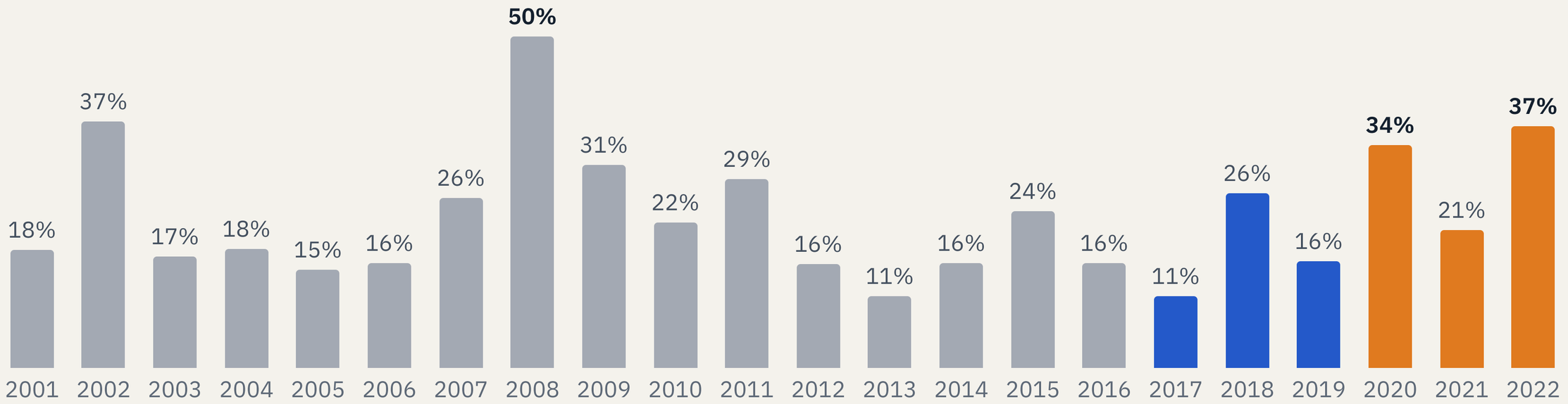
A worse market-wide level forecast cancels the within-month gain. Hourly volatility does not explain it. It needs confirming on new data.



Test leaderboard: all candidates overlap within their CIs; the model chosen on validation (★) sits mid-pack.

The drawdown rate swings from 11% to 50%

Share of stock-days with a 10% drawdown, by year: train validation test



2020: the COVID crash

Volatility R^2 fell to 0.27, against 0.63 in 2021 and 0.57 in 2022. A market-wide shock no single stock's history could foresee.

Calm validation years

2017–2019 held no crisis (17.8% drawdowns vs 30.2% in test), so selection could not reward crisis skill.

Four biases we name rather than hide

Survivorship

The universe holds only stocks that still trade, so risk is probably understated for firms that later failed.

Base-rate shift

The drawdown rate rose from 17.8% in validation to 30.2% in test, and calibration error rose to 0.058.

Selection luck

Picking from many models on one 3-year window rewards noise: the chosen drawdown model ranked last of 17 on test, all within 0.012 AUC.

Reused test years

Earlier runs read 2020–2022, so the test is a historical check. A clean test needs data after October 2026.

One metric needs care: direction’s F1 of 0.67 at recall 1.00 is what flagging every stock scores, not skill.

Designed against misuse, open about limits

Misuse risk

- NO TRADE is the default decision
- Direction calls stay hidden: no edge
- Plain-language explanations and risk tiers
- Labelled research output, not investment advice

Uneven errors

- Errors peak in crises, when users most need accuracy
- Survivorship may make fragile firms look safer
- Calibration helps; it does not replace judgement

Transparency and cost

- Code, reports and metric provenance are public
- Data and weights stay private: licence not cleared
- About 4.5 A100-hours for the whole run
- Uploaded screenshots are read in memory, never stored

AI use: AI assistants (mainly Claude Code, plus Antigravity and ChatGPT) wrote most of the code, proposed much of the design, ran analyses and drafted the text. The author takes responsibility. Details: Appendix B of the paper.

CONCLUSION

What we learned, and what comes next

ESTABLISHED

- Risk is predictable: volatility R^2 0.503, drawdown AUC 0.648
- Ahead of classical models on squared error, though not significantly over HAR
- Fine-tuning helps only when the whole network adapts
- Convex heads certified at their global minimum
- Images: no gain in overall R^2, +0.020 within months (found after the fact); no 5-day edge

NEXT

- Confirm the within-month gain on data after October 2026
- Walk-forward evaluation that includes 2008 and 2020
- Average the top networks instead of one winner
- A probability floor and rolling recalibration
- A research-grade data set that includes delisted firms

Thank you · github.com/harshatangirala/triframe-cv